MILITARY SPECIFICATION

ENGAGING DEVICE, AIRCRAFT ARRESTING BARRIER BAK-11/F

1. SCOPE

1.1 This specification covers one type of aircraft arresting barrier engaging device, designated BAK-11/F.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-P-6906	Plates, Identification
MIL-C-9002	Compressor, Power Driven, Aircraft, Electric Motor Driven, 15 CFM, 3,500 PSI, Type MB-1
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Military

MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-803	Human Engineering Design Criteria for Aerospace Systems and Equipment (Part I Aerospace Systems Ground Equipment)
MIL-STD-808	Finishes, Protective, and Codes, for Ground and Ground Support Equipment
MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment
MIL-STD-831	Test Reports, Preparation of

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DRAWINGS

Air Force

61D1220	Valve - Rapid Opening, Assembly of
61D1248	Trough, Cable - Aircraft Arresting Barrier, Assembly of

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society of Mechanical Engineers

Power Test Code

Unfired Pressure Vessels Code

(Copies of ASME Power Test Codes may be obtained from the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York 17, New York.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Components. The engaging device shall consist of the following major components:

Item or System Name	See Requirement
Electronic control system	3.7
Timing switch mats	3.8
Cable trough	3.9
Pneumatic system	3.10
Interconnect system	3.11
Arresting pendant	3.12.

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3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials. Materials shall be as specified herein. Materials not definitely specified shall be of the best commercial quality and entirely suitable for the purpose. Materials shall be free from all defects and imperfections that might affect the serviceability of the finished item.

3.5 Design. The engaging device shall be an electronic, pneumatic system which may be combined with any standard USAF energy absorber to form a complete aircraft arresting barrier. The engaging device shall be so designed that it will engage any overrunning aircraft under emergency conditions. The emergency engagement shall be accomplished by detecting the approaching aircraft and ejecting an arresting cable from below the runway surface to intercept the main landing wheel struts above the tires. The detection of the approaching aircraft shall be accomplished by monitoring responses to the aircraft rollover of the runway-mounted switch mats.

3.5.1 Aircraft data. The engaging device shall be designed to engage all USAF aircraft without requiring special information input regarding the aircraft type or external store configuration. The engaging device shall safely engage aircraft at speeds from 10 to 180 knots at off-center distances as far as 30 feet from the runway edge. The engaging device shall be so designed that it can be recycled within 5 minutes and operated 12 times in any 24-hour period. The device shall be capable of distinguishing between aircraft having a wheelbase greater or smaller than 25 feet and causing the energy absorber to appropriately select high or low weight setting on adjustable absorber units.

3.5.2 Trailing hook aircraft. The engaging device shall be designed for use in conjunction with the standard supported arresting pendant used in the arrestment of aircraft equipped with a trailing arresting hook. The arresting cable of the engaging device shall be disconnected from the energy absorber in the event of a successful hook engagement of the supported arresting cable (see figure 1). In addition, the engaging device shall not attempt an engagement against an aircraft which has accomplished a successful arresting hook engagement against the supported cable.

3.5.3 Stub hook aircraft. The engaging device shall be so designed that it can be programmed to eject the arresting cable to engage a fuselage-mounted stub-type arresting hook located aft of the main landing gear. The engagement of a stub arresting hook by the engaging device shall be considered to be an auxiliary mode of operation and

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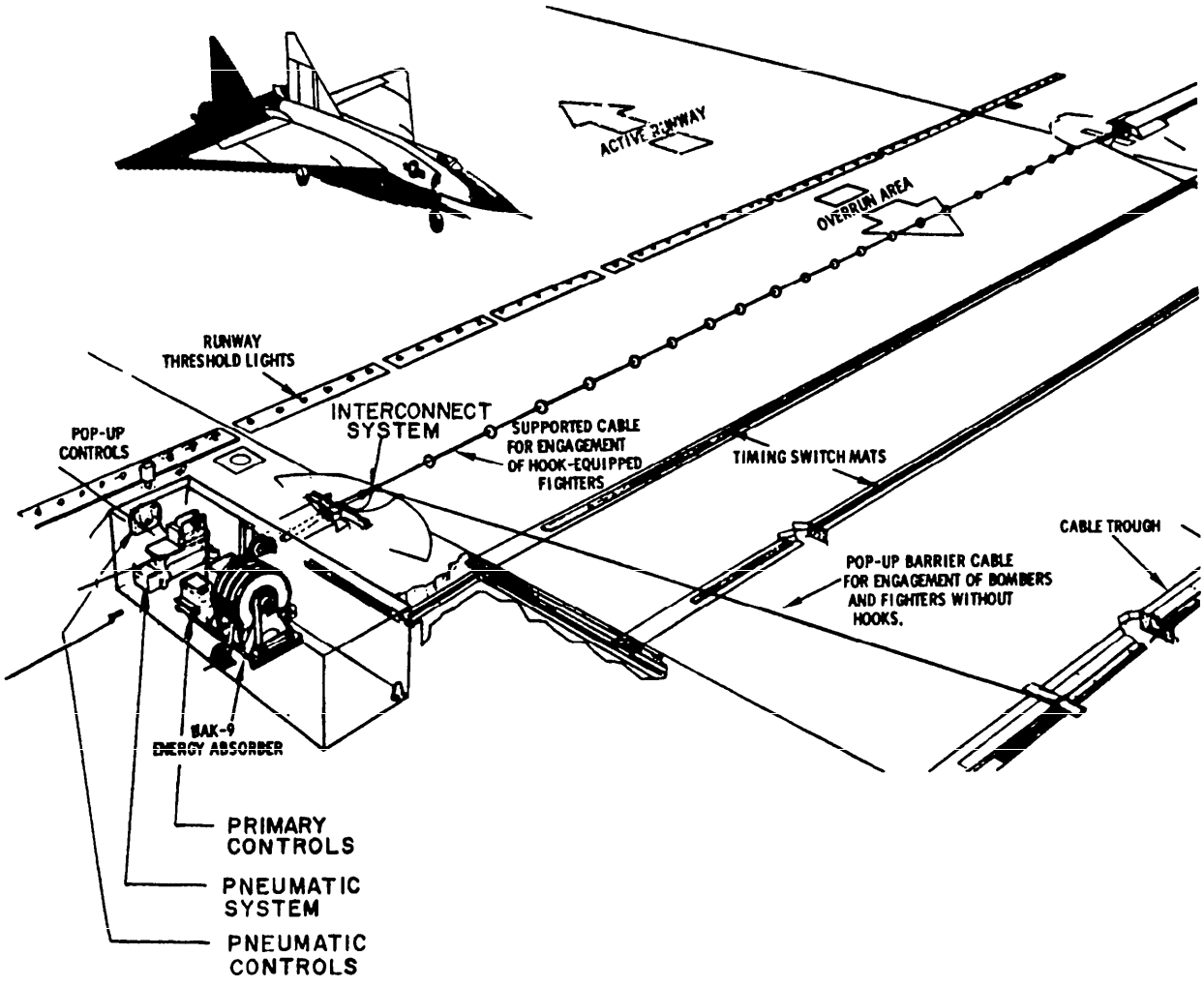


FIGURE 1. Engaging Device Aircraft Arresting Barrier BAK-11/F

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shall not jeopardize the primary function of the engaging device which is emergency engagement of aircraft without a usable trailing arresting hook.

3.5.4 Safety. Rotating and reciprocating parts that could constitute a hazard to operating personnel shall be fully enclosed or properly guarded. Exposed parts that are subject to high operating temperatures or that are electrically energized shall be insulated, fully enclosed, or adequately guarded.

3.5.5 Reliability. The engaging device shall have a minimum mean-time-between failure (MTBF) of 720 hours with a minimum reliability of 96.7 percent for a 24-hour mission at a confidence factor of 0.90 (see 6.3).

3.5.6 Maintainability. The engaging device shall be designed and constructed as specified herein and to provide the following:

a. Minimum number of parts consistent with reliability and performance specified herein

b. Minimum amount of training and time necessary for assembly, disassembly, location of trouble sources, and maintenance including servicing. Where practical, parts and components shall be located or positioned for rapid and simple inspection and recognition of excessive wear or potential failure

c. Permit adjustments, servicing, replacement of parts and components, and other maintenance with minimum disturbance to other equipment parts or components. Parts and components shall be located for ample and rapid access unless performance or reliability will be appreciably degraded by the accessible location. In performing maintenance, if engineering reasoning or data determines that physical or visual interference between items cannot be avoided, the item predicted to require the most maintenance shall be located for best accessibility

d. Permit maintenance with general-purpose tools and equipment normally available commercially. Use of special-purpose tools and equipment shall be subject to approval by the procuring activity

e. Minimum number of tools required for maintenance by design practices such as reducing the variety of bolt head sizes to the practicable minimum

f. Foolproofness - Where improper installation of an item could cause malfunction of the item or the equipment in which it is installed,

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an unsymmetrical mounting means shall be provided where practical. That mounting shall allow the item to be installed only in its proper operating position. If an unsymmetrical mounting means is not practical, the item shall be so mounted that its proper operating position can be readily and visibly determined by service maintenance personnel.

3.5.7 Installation. All engaging system components shall be designed to be installed flush with the surrounding terrain. The switch mats and cable trough shall be flush with the runway. All controls and pneumatic systems shall be installed in a pit(s) and shall have a cover(s) that will withstand aircraft wheel loads comparable with the surroundings.

3.5.8 Environmental. The engaging device shall be designed to operate satisfactorily in ambient temperatures ranging from -40° to +125°F, and in all conditions of humidity including rain.

3.5.9 Lightning. The engaging device shall be protected from overvoltages induced by lightning.

3.6 Construction. The engaging device shall be so constructed that no part will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, and service.

3.7 Electronic control system

3.7.1 Primary control system. The engaging device shall be capable of receiving electrical signals generated by the aircraft wheels passing over the runway-mounted switch mats described in 3.8, and causing the rapid opening valves (see 3.9.1) to open at the correct instant so as to affect engagement of the overruning aircraft. The aircraft shall be engaged by an arresting cable which is propelled vertically upward from below the runway surface by the action of the high pressure air escaping from the rapid opening valves. The vertically rising cable shall intercept the passing aircraft at the main landing wheel struts above the aircraft tires. The lead time (see 6.3) shall be continuously adjustable to account for all possible aircraft geometry situations which may arise. Valve actuation shall be timed to an accuracy of ±0.0005 second for aircraft speeds from 20 to 180 knots. Electromechanical and semiconductor control systems are acceptable.

3.7.1.1 Operational mode capability. Provisions shall be made to allow reprogramming of the engaging device to affect engagement of a fuselage-mounted stub-arresting hook. Lead time for operational mode engagements shall be continuously adjustable to accommodate

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hook locations from 3 to 20 feet aft of the main landing gear. The reprogramming of the engaging device shall be readily accomplished from the main control panel only.

3.7.1.2 Segmenting controls

3.7.1.2.1 Switch mat segmenting. The control system shall be programmed to respond to the pulses generated by the switch module(s) contacted by the nose wheel of the overrunning aircraft and the two adjacent modules on each side. Signals received from switch modules other than these five or six switch modules shall be disregarded. In this manner, signals received by the contact of outrigger wheels of bomber aircraft or emergency vehicles in chase of the overrunning aircraft will not affect the engaging device.

3.7.1.2.2 Valve segmenting. The rapid opening valve nearest the aircraft line of travel and the two adjacent valves on each side of that valve shall be actuated to cause the cable to be ejected from the runway slot. In the event the aircraft line of travel is equidistant from two valves, a total of six valves may be actuated. In this manner, only that portion of cable directly in front of the overrunning aircraft shall be ejected from the runway slot.

3.7.1.2.3 Segmenting control modules. Each segmenting module shall consist of one switch mat module in each of the two timing switch rows and one rapid opening valve. The centers of the switch mats and the rapid opening valve shall be in line and equidistant from the runway centerline.

3.7.1.2.3.1 Segmenting control module indicator lights. Each segmenting control module shall contain two indicator lights. The lights shall be mounted on a console in the control set. Each light shall be normally off. One light shall indicate pneumatic pressure at the rapid opening valve. When the pneumatic pressure drops to 85 percent of the normal operating pressure, this light will so indicate by illumination, and remain illuminated until the proper pressure is restored. The other light shall indicate when that module has been alerted and is ready to fire.

3.7.1.3 Minimum speed cutout. The engaging device shall not attempt an engagement against an aircraft which enters the barrier area at a low velocity. Selection of the low velocity shall be left to the discretion of the using activity. Provisions shall be included, on the main control panel only, to program the minimum speed cutout. The minimum speed cutout control shall be readily accessible and continuously variable between 10 and 50 knots. Stops shall be incorporated into the minimum

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speed cutout to preclude setting of less than 10 knots or more than 50 knots. After minimum speed cutout actuation, automatic reset after 5 to 10 seconds delay shall reposition the engaging device into the ready condition.

3.7.1.3.1 Reset

3.7.1.3.1.1 Automatic reset. In the event that an incomplete set of pulses are received from the runway switches, the engaging device shall automatically reset itself. The reset shall be delayed between 5 and 10 seconds.

3.7.1.3.1.2 Manual reset. The engaging device shall require manual reset after any cable ejection. Manual reset may be accomplished at any previously assigned control station.

3.7.1.4 Fail safe system. To preclude nose wheel engagement, the engaging device shall include a switch row located at the cable slot. The cable ejection shall, in all instances, be delayed until the nose wheel of the overrunning aircraft has contacted this switch. The cable trough switch need not be segmented but must display the same mechanical and electrical properties of the timing switch modules.

3.7.1.5 Supported pendant cutout. When used in conjunction with a supported arresting pendant, the engaging device shall not attempt an emergency engagement against a hook-equipped aircraft which has successfully engaged the supported pendant. Manual reset of the engaging device shall be required after a successful hook engagement has disarmed the engaging device. The supported pendant cutout shall be a positive, mechanical, disconnect-type switch which is actuated to disarm the engaging device upon a predetermined movement of the supported arresting pendant. This movement of the supported pendant shall be sufficient to assure a positive hook engagement has been accomplished.

3.7.1.6 Aircraft identification. USAF jet aircraft geometry is such that all fighter type aircraft have a wheelbase smaller than 25 feet and bomber/cargo type aircraft have a wheelbase longer than 25 feet. The difference in aircraft geometry shall not confuse the timing circuits of the engaging device. The engaging device shall be capable of recognizing this difference and signalling the energy absorber accordingly.

3.7.1.7 Malfunction indication. All engaging device circuits shall be automatically monitored. Failure of any portion of the engaging device

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shall be indicated on all control consoles as a malfunction. The main control console shall be fitted with individual warning lights which will indicate the failures in the following systems:

a. Low air pressure at cable trough

b. Supported pendant cutout

c. Automatic reset

d. Runway switch mat row (individual light for each row)

e. Firing circuit

f. Segmenting controls

g. Power input.

3.7.2 Checkout facilities. Provisions shall be incorporated into the control system to checkout the aircraft timing circuits, the segmenting circuits, and the rapid opening valves. Other control circuits shall be checked out as a by-product of the above mentioned procedures. All test controls shall be located on the main control console as specified in 3.7.3.1. The engaging device shall not be ready for an aircraft engagement during the checkout periods.

3.7.2.1 Timer checkout. Timer checkout shall be accomplished by switching the control system from the ready state to timer test state. Two momentary contact, normally open, pushbutton switches shall be used to simulate aircraft wheel contacts of the timing switch rows. By proper actuation of the pushbuttons, aircraft rollover may be simulated. A clearly noticeable (audible or visual) signal shall indicate proper timer functioning.

3.7.2.1.1 Minimum speed cutout check. The minimum speed cutout shall be checked using the timer checkout facility. Proper timing of the actuation of the pushbuttons may simulate an aircraft entering the barrier area below the preset minimum speed. Automatic reset after 5 to 10 seconds shall indicate proper functioning of the minimum speed cutout.

3.7.2.1.2 Automatic reset check. The automatic reset capability of the engaging device shall be checked by simulating an incomplete signal sequence via the pushbuttons. Automatic reset after 5 to 10 seconds delay shall indicate proper functioning.

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3.7.2.1.3 Aircraft identification checkout. The aircraft size identification circuit shall be checked using the timer test facility. The different sequence of aircraft wheels contacting the timing switch row can be simulated using the two pushbuttons. The signal used in the timer check combined with the actuation of the BAK-9/F48A energy absorber weight selector shall indicate proper functioning of the aircraft identification circuit.

3.7.2.2 Valve check. The individual rapid opening valves shall be test fired by switching the control system from the ready state to the valve test state. Provisions shall be made to select any rapid opening valve from the main control panel and firing this valve by depressing a momentary contact, normally open, pushbutton switch.

3.7.2.3 Segmenting control checkout. The segmenting control checkout shall be accomplished by switching the control system from the ready state to the segmenting control checkout state. The automatic reset shall be turned off. The segmenting circuits shall be checked by manually actuating one switch mat in the first row of timing switch mats and checking segmentation on the segmenting control module indicator lights (see 3.7.1.2.3.1).

3.7.2.3.1 Multiple valve check. With the control system arranged for the segmenting control check (3.7.2.3) and the proper five segmenting modules selected (3.7.1.2.2), the five valves shall be fired by switching the control system to valve check (3.7.2.2) and pushing the fire button.

3.7.3 Control stations. The engaging device shall have three control stations, each designed in accordance with MIL-STD-803, and located as specified herein.

3.7.3.1 Main control station. The main control station shall be located in a concrete vault at the runway edge in close proximity with the primary control system and pneumatic system (see figure 1). It shall consist of one panel so positioned as to be readily accessible to a man standing on the floor of the pit. The panel shall contain the following controls and indicator lights:

a. ON-OFF switches and indicator lights

b. Malfunction indicator lights

c. Lead time adjustments

d. Minimum cutoff speed adjustment

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e. Automatic reset ON and OFF switch

f. Manual reset button

g. Emergency or operational mode selection

h. Local or remote control selector switch and indicator

i. Checkout selector switch

j. Two timer checkout pushbuttons

k. Valve checkout selector switch

l. Valve firing button

m. Adjusting and tuning connectors, if required.

3.7.3.2 Runway edge control station. The runway edge control station shall be surface mounted at the runway edge, positioned so as to provide access from ground level. Anyone operating the control station shall have an unobstructed view of the entire barrier area, overrun, and runway. Any above ground installation shall be so mounted as to afford a minimum obstruction to any vehicle which may strike it. The control and indicator lights shall be encased in a weatherproof container with a hinged cover that can be locked. The control panel shall contain the following controls and indicator lights:

a. ON-OFF switches and indicator lights

b. Malfunction indicator light

c. Runway edge or control tower selector switch and indicator light.

3.7.3.2.1 Warning light. A red warning light shall be positioned on the runway edge control station and shall be clearly visible from any location within 100 yards of the control station in bright daylight. The warning light shall be illuminated to indicate that the engaging device is in ready state and should be approached with caution.

3.7.3.3 Control tower station. The contron tower station shall be located in the control tower as part of the tower operator's auxiliary console. In the event more than one engaging device is deployed at a single airbase, each control tower station shall be clearly marked as to which end of which runway is concerned. Switches and indicator

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lights shall be provided to control and indicate ON and OFF. The control tower panel shall also indicate an engaging device malfunction and tower or runway edge control. The engaging device malfunction shall be indicated by a flashing red light. All switches and pushbuttons located within the control tower shall be guarded.

3.7.3.3.1 The control tower station shall be designed to accommodate tower control and monitor of the engaging device via buried control cable or two-way fixed frequency radio (see 6.2). Radio control may be used in the event (1) no control cable is in existence at the time of installation of the BAK-11/F and (2) the radio control is economically and technically superior to providing a suitable control cable. The radio control shall meet applicable military standards and shall be shielded from interference from any external source. Coded signals shall be used for control commands. A usable frequency shall be authorized prior to equipment installation.

3.7.3.4 Segmenting control indicator lights console. All segmenting control indicator lights shall be located on a single console in close proximity with the main control panel. This console shall remain covered, except during checkout operations. Each group of indicator lights referring to a single segmenting module shall be arranged together and clearly marked according to module number. Each indicator light shall also be marked as to function. A circuit shall be included in the console to facilitate checking the indicator lights. A single, clearly marked, pushbutton switch shall be provided which, when depressed, shall illuminate all indicator lights located on the console.

3.8 Switch mats. The runway switch mats shall be pressure-sensitive strip switches 6 feet long. The switch mats shall be mounted flush with the runway surface in rows perpendicular to the runway centerline. The number of switch mats per row shall be governed by the runway width (see 6.2). The switch rows shall extend to within 25 feet of the runway edge. Each switch mat shall be so installed as to be readily accessible for any necessary maintenance or inspection. The switch mats shall be hermetically sealed by a material which is nonconductive in all practicable environments. Adequate drain provisions shall be made to prevent the accumulation of ground water at the switch rows. The switch rows shall not obstruct or be damaged by normal operation of snow removal and runway cleaning equipment. In addition, deployed arresting hooks shall not cause damage to, or be deflected by, the switch rows. Each switch mat shall respond to a direct pressure of 14 psi or more and shall withstand the loads imposed by rollover of any aircraft in the USAF inventory.

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3.9 Cable trough. The cable trough segments shall be constructed in accordance with Drawing 61D1248 (see 6.2). In addition, the arresting cable slot shall be adjustable to accommodate arresting pendant diameters from 3/4 to 1-1/2 inches. Each end of the cable trough shall be fitted with an inert end trough which shall be dimensionally the same as the cable trough used throughout the system, except all rapid opening valves and their related hardware shall be removed leaving only the electrical harness and the pneumatic supply lines. The end troughs shall be of equal length for each side of the runway (see 6.2).

3.9.1 Rapid opening valves. The rapid opening valves shall be constructed in accordance with Drawing 61D1220. Provisions shall be incorporated for electrically heating the valves to prevent freezing and malfunctioning of the valve assemblies when adverse environmental conditions prevail.

3.10 Pneumatic system

3.10.1 Compressor. The compressor shall conform to MIL-C-9002.

3.10.2 Receivers. Air receivers shall be provided to meet the recycle requirements specified in 3.5.1 and shall be in accordance with MIL-C-9002.

3.10.3 Pressure warning switch. A pressure warning light switch shall be incorporated into the pneumatic system so as to be compatible with the requirements herein.

3.10.4 Safety factor. All pressure vessels shall be fabricated in accordance with good engineering practices, and shall have a safety factor of not less than 4 to 1 based on the yield strength of the material as set forth in the ASME Code for Unfired Pressure Vessels.

3.11 Interconnect system. The engaging device shall be adaptable for use in conjunction with the supported cable used for the arrestment of hook-equipped aircraft (see figure 1). In the event of a successful hook-supported cable engagement, the engaging device cable shall be disconnected from the supported cable so as not to interfere with arrestment. In addition, the engaging device shall be disarmed in the event of a successful hook engagement (see 3.7.1.5). If the engaging device is actuated, the cable shall be securely locked to the energy absorber.

3.11.1 Pretensioning mechanism. When required (see 6.2), provisions shall be made to pretension the engaging device cable to restrict the

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rise height as required. The pretensioning mechanism shall be a coil spring type at each side of the runway and anchored to suitable concrete anchors.

3.12 Arresting pendant. The arresting pendant shall be 1- or 1-1/4-inch diameter improved plow steel, preformed nonrotating wire rope with independent wire rope center (see 6.2). The 1-inch cable shall have a minimum breaking strength of 84,000 pounds and the 1-1/4-inch cable shall have a minimum breaking strength of 120,000 pounds. The 1-inch cable shall be used for installations incorporating single energy absorbers (BAK-9 or BAK-12). Dual absorber installations require the 1-1/4-inch arresting pendant.

3.13 Electrical requirements. Standby power requirements for the electronic control system shall not exceed 600w. Power required during valve operation shall not exceed 3,000w for 0.030 second. Electrical power supplied to the engaging device shall be 220V, 3 phase, 4 wire, 60 cycle.

3.14 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-D-70327 shall govern the manufacturer's part numbers and changes thereto.

3.15 Interference. Interference control requirements shall be in accordance with MIL-I-6181.

3.16 Finishes and protective coatings. Cleaning, painting, plating, chemical films, and anodic coatings shall be in accordance with MIL-STD-808. All painted surfaces shall be painted for type I exposure. The colors shall be as follows:

a. Exterior surfaces - Yellow, designation DG

g. Interior surfaces - Aircraft gray, designation LG

c. Equipment within - Aircraft gray, designation LG
enclosure (engine,
pumps, et cetera)

3.17 Operational markings

3.17.1 Control markings. All valves, gages, controls, and indicators used in the operation of the engaging device shall be identified by securely attached nameplates conforming to MIL-P-6906. The nameplate

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shall be of such composition that exposure to oil, dirt, light, et cetera, will not cause them to fade or become eradicated. Tags shall not be used.

3.17.2 Operational instructions. Where applicable, brief operating and precautionary instructions for the engaging device shall be permanently affixed on or near the components involved. The instructions shall be clear, concise, and adequate to enable operation without damage to the unit or injury to operating personnel. Instruction panels shall be of sheet aluminum or sheet zinc not less than 0.050 inch thick, engraved or etched to produce raised or depressed markings with a black or other dark color background, and with a border of not less than 1/4 inch.

3.18 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.19 Workmanship. The engaging device, including all parts and accessories shall be fabricated and finished in a workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring and riveting; alignment of parts and tightness of assembly screws and bolts.

3.19.1 Riveting. Riveting operations shall be carefully performed to insure tightness and satisfactory heading of rivets.

3.19.2 Cleaning. The engaging device shall be thoroughly cleaned and loose, spattered, or excess solder, metal chips, and other foreign material removed during and after final assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the engaging device shall be classified as follows:

a. Preproduction testing . . . . . See 4.4

b. Acceptance tests . . . . . See 4.5.

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4.3 Test conditions

4.3.1 Preparation for test. The engaging device submitted for test shall be correctly adjusted, lubricated, and serviced for immediate operation.

4.4 Preproduction testing (see 6.2)

4.4.1 Test sample. One engaging device shall be subjected to the tests specified in 4.4.3.

4.4.2 Test report. After completion of the preproduction tests, a test report shall be prepared in accordance with MIL-STD-831 and three complete copies of the report furnished the procuring activity.

4.4.2.1 Reliability and maintainability information. The following information shall be included as an appendix to the test report:

a. All failures, servicing, adjustments, maintenance, and irregular functioning shall be identified by accumulated operating time, cycles, miles, or position in the test procedure, as appropriate. Test conditions at the time of the events identified shall be recorded

b. A summary of the engineering analysis and of any tests conducted to determine assignable causes for any failure or irregular functioning

c. A summary of the engineering analysis leading to any corrections made to design, construction, quality control, or other procedures, or leading to any corrections to be made to production items or proposed to be made. The summary shall also include an analysis of the predicted effectiveness of such corrections

d. Clock time and man-hours required for each maintenance and servicing action taken during the tests. A brief description of the experience and qualifications of the personnel taking such actions shall be included

e. Test activity or contractor comments on item features or requirements that, if modified, should improve the item

f. Test activity or contractor comments on use or maintenance conditions to be avoided or cultivated to increase the reliability or useful life of the item

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g. Any of the above information that is already included in the test report body need not be repeated in the information required by this paragraph, but clear reference to the location of the data shall be included.

4.4.3 Preproduction tests. The preproduction tests shall consist of all tests specified under 4.6.

4.5 Acceptance tests. The acceptance tests shall consist of the following:

a. Individual tests

b. Sampling plan and test.

4.5.1 Individual tests

4.5.1.1 Uninstalled engaging device. Each uninstalled engaging device shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Mechanical inspection

c. Electronic inspection

d. Component tests

e. Submersion test.

4.5.1.2 Installed engaging device. Each installed engaging device shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Mechanical inspection

c. Electronic inspection

d. Control station

e. Functional check

f. Endurance test

g. Submersion test.

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4.5.2 Sampling plan and test

4.5.2.1 Switch mat lots. All production quantities shall be divided into lots of one hundred consecutively produced switch mats.

4.5.2.1.1 Switch mat sampling and test. Ten switch mats from each lot shall be selected at random and subjected to the submersion test specified in 4.6.8.1, except the test duration shall be 36 hours and the temperature cycles shall be not less than 4 hours nor more than 6 hours.

4.5.2.1.2 Switch mat rejection and retest. The following criteria shall be used as a basis for acceptance of the switch mat:

a. No failures in first sample of 10 items - accept entire lot

b. One failure in first sample of 10 items - reject failed item and test 10 additional items

(1) No failure in additional sample - accept lot

(2) One failure in additional sample - reject failure and test second additional sample of 10 items

(3) One or more failures in second additional sample - reject failed items and test entire lot

c. Two or more failures in first sample of 10 items - reject failed items and test entire lot. All failed items shall be rejected regardless of status of lot.

4.6 Test methods

4.6.1 Examination of product. The engaging device shall be visually examined for compliance with the requirements of this specification with regard to completeness of assembly, presence of all components and accessories, quality of workmanship and materials, dimensions, hydraulic component leaks, marking, mounting suitability and simplicity, and the functional operation of all components, assemblies, accessories, and controls.

4.6.2 Mechanical inspection. A critical mechanical inspection shall be made of all components and parts of the engaging device to insure proper functioning of all mechanical parts, accessibility of all controls and adjustments, installation and interchangeability of removable

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parts without special fitting, and to effect every degree of safety and performance. The inspection shall also include the following:

4.6.2.1 Valve test. Each valve shall be fired to test functioning and reset capability. Thirty seconds after each valve is fired, it shall be refired. After each valve has been fired two times, the entire pneumatic system shall be checked for leaks and proper manifold pressures.

4.6.2.2 Cycle rate test. Six adjacent valves shall be fired 12 times at 5-minute intervals to inspect the system cycle capabilities. Cable rise height and time may be used as an indication of system performance.

4.6.2.3 Pretension test. When the pretension mechanism is specified, it shall be tested to determine what degree of pretension will be required to obtain the desired cable rise height. The cable rise height shall not exceed 6 feet nor be less than 3 feet at the center of the thrown section.

4.6.3 Electronic inspection. A critical electronic inspection shall be made of all components of the engaging device. The inspection shall include any servicing to insure readiness for operation if required.

4.6.3.1 Switch mat inspection. All switch mat modules shall be critically inspected for any damage. All modules shall be checked for any closed or shorted switch mats. All defective modules shall be replaced.

4.6.4 Control stations. All control stations shall be checked for compliance with the requirements specified herein. Three-wire control as well as fixed frequency radio control via control tower station shall be tested.

4.6.5 Functional check. When specified at the installation site, the contractor shall functionally test the engaging device. This test shall consist of driving an appropriate vehicle (not necessarily an aircraft) over the engaging device (with the cable removed from the slot) at various speeds to determine proper system functioning.

4.6.6 Functional test (to be conducted by the procuring activity). The engaging device shall be functionally tested against manned aircraft to determine compliance with the requirements herein. The test shall be made up to the maximum speed safely obtainable at the test installation and shall be made on-center and off-center as far as 60 percent of the semispan of the runway.

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4.6.7 Component tests

4.6.7.1 Cable trough segment. The cable trough segment shall be tested as follows to determine compliance with the requirements specified herein. Compressed air and electrical energy shall be supplied to the cable trough segment which closely simulates the installed conditions. Each valve shall be fired to test functioning and reset capability. Thirty seconds after each valve is fired, it shall be refired and shall perform equal to the first firing.

4.6.7.2 Pneumatic system. The compressor, air storage system, and all pneumatic controls shall be tested to determine compliance with the requirements specified herein. The pneumatic system shall be installed in a test stand which closely approximates the installed conditions and operated to assure proper functioning of all components.

4.6.7.3 Control system. The control system shall be tested to determine compliance with the requirements specified herein. The control system shall be installed in a test stand which closely approximates the installed conditions. A mechanism which can supply signals to the control system to simulate the signals generated by aircraft rollover of the switch mats at various speeds and wheel spans, both with or without outrigger wheels, shall be used. Compliance with the requirements shall be determined from oscillograph traces of the control system output.

4.6.8 Environmental tests. The switch mats shall be subjected to the following tests conducted in accordance with the specified methods of MIL-STD-810:

a. High temperature - Method 501.1, procedure I

b. Low temperature - Method 502.1, procedure I

c. Humidity - Method 507.1, procedure I.

4.6.8.1 Submersion test. All switch mat modules shall be submerged in distilled water having a pH value between 6.5 and 7.5 at 77°F for a period of 10 days. The water temperature shall be cycled between 40° and 120°F in cycles not less than 6 hours nor more than 10 hours duration. After 10 days of submersion, the switch mats shall be checked for electrical resistance. The electrical resistance shall not vary more than ±20 percent of the unsoaked switch mat.

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4.6.9 Endurance test (to be conducted by the procuring activity). The installed engaging device shall be operated in the standby condition for a total of 60 days without a failure (see 6.3.2). The endurance test shall be operated on a basis of 24-hour mission time with normal inspection and maintenance being performed at the end of each mission (see 6.3.3). The maintenance and inspection time shall be included in the mission time. Near the end of each mission and prior to the maintenance and inspection, the engaging device shall be subjected to a simulated engagement.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with the instructions of the procuring activity.

6. NOTES

6.1 Intended use. The BAK-11/F engaging device is intended to accomplish engagement of nonhook-equipped aircraft penetrating the runway overrun at speeds ranging from 10 to 180 knots. The engaging device must be used with a suitable braking unit (energy absorber) to provide a complete arresting system.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Switch mat segments per row - 21, 29, or 36 (see 3.8)

c. Cable trough segments - 12, 14, or 23 (see 3.9)

d. End troughs (each end; 12-foot lengths) - 1-1/2, 1-1/3, or 1 (see 3.9)

e. Runway width - 150, 200, or 300 feet

f. Pretension - as required (see 3.11.1)

g. Desired cable rise height (see 3.11.1)

h. Length and diameter of arresting pendant (see 3.12)

i. Radio control or buried cable (see 3.7.3.3.1)

j. Extent of contractor support required during preproduction test program conducted by the procuring activity (see 4.6.6).

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6.3 Definitions

6.3.1 Reliability. Reliability is the probability of performing without failure a specified function under given conditions for a specified period of time. Recognizing that, in general, the rate of failure of equipment is fairly constant throughout the life of the equipment, the probability of nonfailure over an operating time interval decreases exponentially as a function of the length of the interval, during which time there is a constant failure rate, and can be expressed as follows:

Confidence level: 90 percent

$$\text{MTBF} = \frac{\text{Total test time}}{2.3}$$

$$\text{Reliability} = e^{-x} \quad \text{where } x = \frac{\text{Mission time}}{\text{MTBF}}$$

The $2.3 = \frac{4.61}{2}$ is based on the constant for the Poisson/Chi squared distribution, assuming an exponential (2 degrees freedom) distribution even though the failure rate for the test is zero.

6.3.2 Mean-time-between-failure. The mean life or mean-time-between-failure is the arithmetical mean (average) of the operating time between failures.

6.3.3 Normal inspection. Normal inspection consist of firing every rapid opening valve in the engaging device.

6.3.4 Mission. It is the mission of the barrier system to stand ready to accomplish emergency arrestment of aircraft which penetrate the runway overrun area and are in a configuration and attitude compatible with the barrier installation.

6.3.5 Mission time. Mission time covers that period of time between periodic inspections and implementation of routine maintenance procedures.

6.3.6 Failure. Any malfunction which prevents the engaging device from fulfilling the mission or which requires a maintenance operation prior to completion of mission time, shall be considered a failure.

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6.3.7 Lead time. Due to the inability of the cable to rise instantaneously, the cable ejection must be initiated a short time interval prior to the arrival of the engagement point over the cable slot. This time interval is called lead time.

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Air Force - 11

Preparing activity:
Air Force - 11

Reviewers:
Air Force - 11, 82

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SPECIFICATION ANALYSIS SHEET	Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use of the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A
☐ Direct Government Contract ☐ Subcontract

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)	DATE

DD Form 1426

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